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# Nanobubbles drive advanced anaerobic treatment of swine wastewater for efficient methane recovery: Performance gains and multi-pathway enhancement

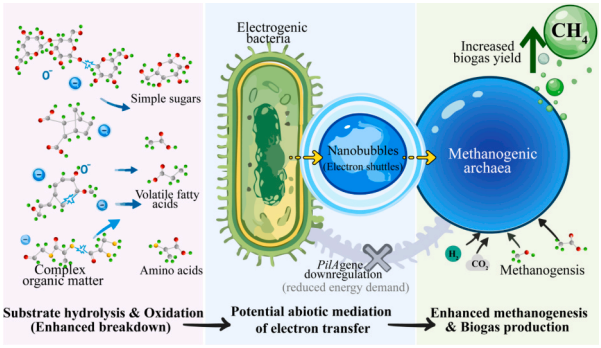
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## HIGHLIGHTS

- $O_3$ nanobubbles boost methane yield from swine wastewater by 87.5%
- Nanobubbles accelerate hydrolysis and acidification in anaerobic digestion.
- Nanobubbles enriched microbes and boosted methanogenic pathways.
- Charged nanobubbles may act as abiotic mediator for electron transfer.

## GRAPHICAL ABSTRACT




## ARTICLE INFO

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## ABSTRACT

Swine wastewater contains recoverable energy, but anaerobic digestion is often limited by complex organics and slow hydrolysis. To overcome this limitation, this study introduced nanobubble technology using three gas media (air, $O_2$, and $O_3$) and systematically studied their effects on methane recovery during the anaerobic digestion of swine wastewater. Batch experiments showed that $O_3$ nanobubbles achieved the strongest enhancement, increasing cumulative methane production by 87.5% compared with the control. This improvement may result from the strong oxidative capacity of $O_3$ nanobubbles to degrade recalcitrant organics, as indicated by the second methane production peak observed only in the $O_3$ nanobubbles. In contrast, $O_2$ nanobubbles provided the weakest improvement, potentially because excess dissolved oxygen stimulated facultative aerobic respiration,

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converting substrates to $CO_2$ and lowering availability for methanogenesis. Further analysis revealed that all nanobubble treatments accelerated volatile fatty acid turnover and enriched key hydrolytic and acidogenic microbes, particularly under $O_3$ nanobubbles. The enrichment of *Methanothrix* and downregulation of the energy-intensive *PilA* gene suggest promoted electron transfer. Negatively charged nanobubbles may act as abiotic mediators that facilitate direct interspecies electron transfer. Metabolic analysis indicated enhanced hydrogenotrophic, methylotrophic, and acetoclastic methanogenesis, implying strengthened synergy among pathways. Overall, $O_3$ nanobubbles show promise for resource recovery from organic waste.

## 1. Introduction

The rapid expansion and intensive development of the swine industry in recent years have led to increased pollution of swine wastewater (SW). Although this type of wastewater contains large amounts of recoverable organic matter, it also harbors toxic substances and pathogens (Zhang et al., 2021). Improper treatment not only wastes biological resources but also pollutes the environment and endangers human health. Therefore, the selection of appropriate methods is critical to SW disposal in an environmental-friendly and cost-efficient way. Anaerobic digestion (AD) represents an efficient swine manure treatment technology capable of simultaneously removing pollutants and recovering resources (Massalha et al., 2020). However, the presence of substances that are difficult to degrade such as cellulose and lignin in SW leads to inefficient hydrolysis, low methane yield, and long digestion time (Li et al., 2020). To enhance the AD efficiency of SW, various strategies have been explored, such as thermal pretreatment, integration with microbial electrolysis systems, and co-digestion (Wang et al., 2020c). However, these methods often have certain limitations, including high energy requirements and complex operation demands (Yang et al., 2019). Therefore, there is increasing interest in developing alternative technologies that are efficient, environmentally friendly, and easily integrated into existing AD systems.

In recent years, nanobubble (NB) technology has emerged as a promising approach for wastewater treatment, with its application scope covering livestock and agricultural wastewater. NBs are tiny bubbles with a diameter of less than 1 μm, possessing unique physical and chemical properties, such as high specific surface area, efficient gas solubility, excellent stability, and prolonged residence times in water (over 14 days) (Wang et al., 2019). NBs have the advantages of high efficiency, low-cost, and the absence of secondary pollution (Yang et al., 2019), and thus are widely used in AD processes. NBs can significantly enhance the gas–liquid mass transfer efficiency and generate free radicals (Yang et al., 2019), thereby effectively promoting the hydrolysis of particulate organic matter, improving microbial substrate accessibility, and stimulating microbial metabolism during the biological treatment. Moreover, NBs can significantly affect the redox process. NBs can produce reactive oxygen species, which help optimize redox potential during denitrification and facilitate sulfide oxidation/reduction to mitigate hydrogen sulfide inhibition. Studies have shown that nanobubble water (NBW) generated from gases such as air, nitrogen, and carbon dioxide can effectively enhance the AD of recalcitrant organic matter (Wang et al., 2023; Wang et al., 2020a). For instance, using cellulose, lignin, and residual activated sludge as substrates, the methane yield in the $N_2$-NBW group increased by 9%, 22%, and 29%, respectively, compared to the untreated group (Wang et al., 2020b; Yang et al., 2019). Among various gases used for NBs generation, $O_3$ has attracted considerable attention due to its strong oxidation capacity and ability to enhance the biodegradability of complex organic substrates. However, conventional $O_3$ processes often suffer from low gas utilization efficiency and high operational costs, primarily due to rapid ozone decomposition and limited mass transfer. Studies have shown that the ozone transfer efficiency in conventional gas bubbling systems is generally below 20–30% (Gottschalk, 2008), necessitating relatively high $O_3$ dosages. Integrating $O_3$ with NB technology offers a potential strategy to overcome these limitations. NB technology generates nanoscale bubbles (diameter $<$ 1 μm) with extremely large specific surface areas and prolonged residence times in water. These characteristics substantially enhance ozone utilization efficiency, enabling effective oxidation at lower ozone concentrations while reducing gas loss and operational energy consumption. Furthermore, NB generation systems do not require complex aeration devices inside the reactor and can alleviate the burden on off-gas treatment systems, thereby simplifying the overall reactor configuration. Previous studies have applied NBs ozonation as a pretreatment step for wastewater treatment. For example, $O_3$ NBs have been used to oxidatively degrade high concentrations of oxytetracycline (Zhou et al., 2024) and to treat ibuprofen-containing pharmaceutical wastewater (Zhou et al., 2025), thereby reducing toxicity and improving the performance of subsequent AD. However, these studies primarily focused on $O_3$ pretreatment prior to digestion, while the direct application of $O_3$ NBs within the anaerobic digestion process remains largely unexplored.

Therefore, this study aims to systematically elucidate the mechanism of action of NBW to promote the performance of AD, with a focus on revealing its microbial and metabolic regulatory mechanisms. This study aims to investigate the effects of NBW generated from different gas sources (air, $O_2$, and $O_3$) on the methane production performance during AD. Secondly, we studied the dynamic changes in the microbial community, focusing on how the addition of NBs reshapes the composition and enriches key functional microorganisms during the hydrolysis, acidification, and methanogenesis. Then, this study elucidates the metabolic coordination promoted by NBs, particularly emphasizing its function as an abiotic mediator facilitating direct interspecies electron transfer (DIET). By combining performance data with metagenomic data, this work provides a comprehensive understanding of the biophysical and biochemical processes driving enhanced methane recovery.

## 2. Materials and methods

### 2.1. Preparation and characterization of nanobubble water

NBW was generated using a NB generator (NANO$^{TM}$ MF-5000, China) following the manufacturer's instructions and a previously reported procedure (Wang et al., 2018). During operation, the liquid flow rate was maintained at 8 L/min and the gas flow rate was set at 0.5 L/h. The gas and deionized water were simultaneously pumped into the generator for 20 min. Three types of gases were applied in this study, including laboratory air, high-purity $O_2$ (99.99%), and high-purity $O_3$ (99.99%). Accordingly, the resulting NBW samples were denoted as Air-NBW, $O_2$-NBW, and $O_3$-NBW, respectively. The particle size distribution and number concentration of NBs in each type of NBW were determined in triplicate prior to use using a nanoparticle tracking analysis system (NanoSight NS500, Malvern Instrument Ltd., UK). To evaluate the stability of NBW, all three NBW samples were stored at ambient temperature for 48 days, during which NB size and concentration were monitored daily.

### 2.2. Batch AD experiment

#### 2.2.1. Raw swine wastewater and inoculum

The raw SW used in this study was collected from a large-scale pig

farm in Beijing, China, and was stored at 4℃ after collection. The SW had a volatile solids/ total solids (VS/TS) ratio of 43.38%, an ammonia nitrogen (TAN) concentration of 453.7 ± 9.3 mg-N/L, a soluble chemical oxygen demand (SCOD) of 3306.7 ± 244.9 mg/L, and a pH of 8.2 (Table S1). The anaerobic inoculum was obtained from a laboratory-scale anaerobic reactor that had been operating stably for an extended period. The inoculum exhibited high stability and low background activity, with the following primary physicochemical properties: a VS/TS ratio of 41.44%, a TAN concentration of 41.3 mg-N/L, a SCOD of 832 mg/L, and a pH of 8.48 (Table S1). The low organic content and mature microbial community of the inoculum ensured minimal endogenous methane production during the BMP assays.

#### 2.2.2. Batch AD test

Batch anaerobic digestion assays were conducted using an AMPTS II system (AMPTS II, Bioprocess Control AB, Sweden) in accordance with the standard biochemical methane potential (BMP) testing protocol. Scott Duran bottles with a total working volume of 500 mL were employed as AD reactors. For each reactor, 242 mL of substrate and 58 mL of inoculum were added, resulting in a VS ratio of 1:1.03 and a TS ratio of 1:1.06, followed by the addition of 100 mL of either deionized water, Air-NBW, $O_2$-NBW, or $O_3$-NBW. The experimental groups are named Air-NBW, $O_2$-NBW, and $O_3$-NBW, with DW (deionized water) used as the control group, and the detailed operating conditions are listed in Table S2. All treatments were performed in triplicate. To establish anaerobic conditions, each reactor was purged with high-purity nitrogen gas (99.99%) for 5 min before sealing. Subsequently, the reactors were incubated in a thermostatic water bath at 37 ± 1℃ for 28 days, with continuous stirring at 120 rpm. Each bottle was equipped with a liquid sampling port and a gas outlet. Liquid samples (10 ml) were withdrawn on days 0, 7, 14, 21, and 28 and stored at −20℃ prior to analysis. The gas outlet of each reactor was connected to an absorption bottle containing 3 M NaOH solution to remove $CO_2$. Methane production was continuously monitored using an online gas measurement system. All methane yields were normalized to standard temperature and pressure (273.15 K and 1 atm) and expressed based on the VS content of the SW added to the reactors.

#### 2.2.3. Physicochemical analysis

TS, VS, total nitrogen (TN) and TAN were determined by standard analytical methods (Fan et al., 2020). pH was determined by a pH meter (SenTix® 940–3, WTW, Germany). For the determination of SCOD, the samples were first filtered through 0.45 μm filter membrane, and then quantitatively analyzed by using Hach reagent tubes. The volatile fatty acid (VFA) was determined by centrifugation, filtration through 0.22 μm filter membrane, and the filtrate was acidified by 5% sulfuric acid, and then analyzed by gas chromatograph (Agilent GC-6890 N, Agilent Technologies, USA) with a hydrogen flame ionization detector and a DB-FATWAX UI column (30 m × 0.25 mm × 0.25 μm, Agilent Technologies, USA). The temperatures of both the injector and detector were set at 250℃, with a split ratio of 25:1. The oven temperature program was as follows: initially held at 80℃ for 1 min, then ramped to 200℃ at a rate of 10 ℃/min. The injection volume was 0.5 μL. Nitrogen was used as the carrier gas at a flow rate of 25 mL/min.

### 2.3. DNA extraction and metagenomic sequencing

The total genomic DNA of the samples was extracted by FastDNA SPIN kit for Soil (MP Biomedicals, USA), and the operation was strictly followed the instructions of the kit. The concentration and purity of the DNA were detected by NanoDrop spectrophotometer (NanoDrop ND-1000, Thermo Scientific, USA), and the integrity of the DNA was verified by 1% agarose gel electrophoresis. Metagenomic sequencing was performed by Majorbio, and the sequencing platform was Illumina HiSeq 2000. In order to construct the paired-end library construction, the genomic DNA was firstly broken to the average fragment length of about 400 bp by Covaris ultrasonicator (M220, Covaris Inc., USA). Then, the library was constructed by using NEXTFLEX™ Rapid DNA-Seq kit (Bioo Scientific, USA), and then amplified by bridge PCR, and then sequenced by using NovaSeq or HiSeq X reagent kits (Illumina, USA). The raw sequence data reported in this paper have been deposited in the NCBI database (PRJNA1438379).

### 2.4. Bioinformatic analysis

The raw sequencing reads were quality filtered by fastp software (https://github.com/OpenGene/fastp) to obtain high quality clean reads. The clean reads were spliced into overlapping clusters by MEGAHIT software (https://github.com/voutcn/megahit), and the minimum length threshold of overlapping clusters was set to 300 bp during the splicing process. Based on the spliced overlap clusters, MetaGene software (https://metagene.cb.k.u-tokyo.ac.jp/) was used to complete the prediction of open reading frames (ORFs). The redundant sequences were clustered using CD-HIT software (https://www.bioinformatics.org/cd-hit/), and the sequence identity and coverage cutoffs were both set at 90%, and the longest sequence in each cluster was selected to construct a nonredundant gene catalog. The nonredundant gene catalog was compared with the NR database and the Kyoto Encyclopedia of Genes and Genomes (KEGG) database, and the functional annotation and species classification annotation of the genes were completed by BLASTP software (version 2.2.28+; E-value $\leq$ 1E − 5). Subsequent annotation analysis and downstream data analysis were completed on the Majorbio BioCloud platform (https://cloud.majorbio.com/). Gene abundance was calculated based on the number of read segments aligned to each gene, and relative abundance was expressed as the ratio of the number of read segments aligned to the target gene to the total number of read segments in a single sample. Annotation and downstream analyses were further performed using the Majorbio Cloud Platform (https://cloud.majorbio.com/). Gene abundance was calculated based on the number of reads mapped to each gene, and relative abundance was expressed as the proportion of reads assigned to a given gene relative to the total reads per sample.

### 2.5. Statistical analysis

Differences in microbial community composition of the samples at the phylum level were analyzed by performing principal component analysis (PCA) in the vegan package of the R software. Heatmaps of the distribution of functional genes involved in each metabolic pathway at different AD stages were done by the online tool ClustVis. The initial processing of the experimental raw data was done using Microsoft Excel 2019 software. Differences in cumulative methane production between treatment groups were tested by one-way analysis of variance (ANOVA) and Tukey HSD, which was analyzed in IBM SPSS Statistics version 22.0, setting $P < 0.05$ as statistically significant difference. Curve fitting and result visualization were achieved by Origin 2019b software. Differences between groups in the relative abundance of functional genes were analyzed by chi-square test, and $P < 0.05$ was used as the criterion to determine that the differences were statistically significant.

## 3. Results

### 3.1. Physicochemical properties of nanobubble water

In this study, NBW was prepared by a NB generator, and the characteristic indexes of the produced NBs are shown in Fig. 1. The size of NBs in the three types of NBW is mainly distributed between 100–300 nm, of which the average diameters of Air NBs, $O_2$ NBs, and $O_3$ NBs are 177.5 ± 15.1 nm, 170.0 ± 31.3 nm, and 171.8 ± 30.2 nm in that order; and the corresponding total number of bubbles of $2.5 \times 10^7$, $2.1 \times 10^7$, and $2.2 \times 10^7$ particles/mL, respectively, which are similar to the results reported in existing studies (X. Wang et al., 2020b). Statistical analysis

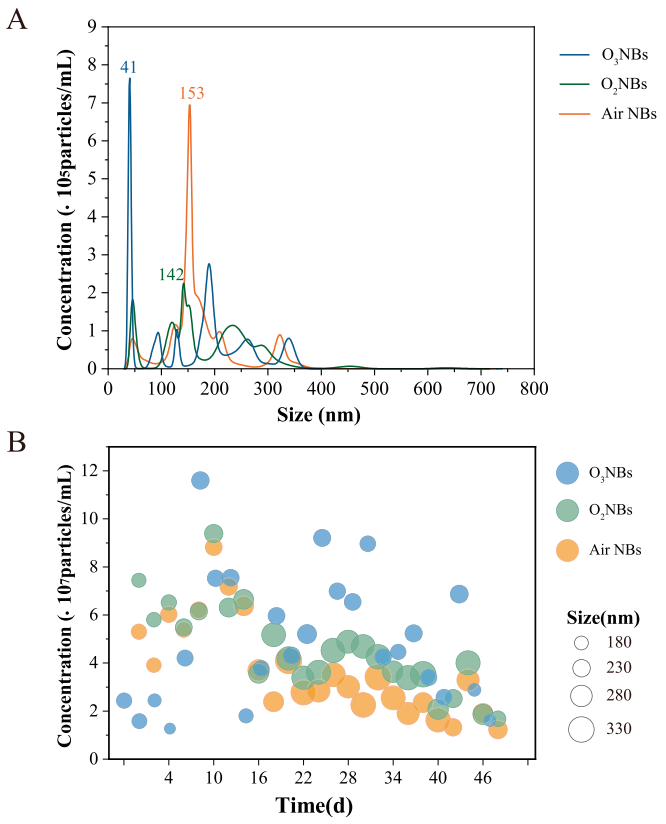



**Fig. 1.** Physicochemical characterization and stability of NBs in aqueous solution. **A** Initial particle size distribution and number concentration of Air, $O_2$, and $O_3$ NBs measured via Nanoparticle Tracking Analysis immediately after generation. The x-axis represents particle diameter (nm), and the y-axis represents particle number concentration. **B** Changes in NB concentration and mean diameter during 48 days of storage. Different colors indicate different gas types: $O_3$ NBs (blue), $O_2$ NBs (green), and Air NBs (orange). In panel B, symbol position indicates NB concentration and storage time, while symbol size represents mean bubble diameter.

showed that there was no significant difference in the average particle size and number of the three NB preparations ($P > 0.05$). This physical consistency ensures comparability between different treatments. Consequently, observed differences in methane recovery primarily stem from the chemical composition and oxidation capacity of the gas, rather than the physical properties of the NBs. This result is consistent with the conclusion of the previous study that the particle size distribution and concentration of NBs tend to stabilize without significant fluctuation when the preparation duration of NBs exceeds 20 min (Wang et al., 2019).

In this study, the retention capacity and stability performance of NBs in aqueous solutions were evaluated by a 48-day storage test, and the changes in bubble particle size and number were monitored (Fig. 1B). The particle size and concentration of Air NB and $O_2$ NB remained relatively stable during the first 14 d of storage; after 14 d, the concentration of the NBs decreased, and the average particle size tended to increase. Even with the above changes, the concentrations of the two types of NBs remained in the order of $10^7$ particles/mL, and the particle sizes were basically in the range of 186–337 nm. In contrast, the concentration of $O_3$ NBs showed fluctuations during the storage period, with values ranging from $2.2 \times 10^7$ to $1.2 \times 10^8$ particles/mL, a phenomenon that may be related to the instability of $O_3$ itself. However, the average particle size of $O_3$ NBs remained relatively stable throughout the storage period, basically around 200 nm (Fig. 1B). The above results demonstrate the successful generation characteristics and storage performance of the prepared NBW in aqueous solution, providing a useful physicochemical basis for its subsequent application in the AD system. However, under the complex digestion matrix, the behavior of NBs, particularly their size evolution and concentration dynamics, remains to be further elucidated using more targeted analytical approaches.

All three NB types showed long-term stability, consistent with reports that NBs can persist in water for over 14 days (Ji et al., 2020; Liu et al., 2016). This stability is mainly explained by their physicochemical features: (i) nanoscale size gives very low buoyancy and a slow rise velocity (Favvas et al., 2021); and (ii) a strongly negative zeta potential (typically −15 to −30 mV) creates an interfacial electric double layer (Varoutoglou et al., 2025), whose electrostatic repulsion inhibits coalescence and collapse (Wang and Lu, 2025). These properties ensured stable NB presence throughout the AD tests.

### 3.2. The influence of nanobubble water on methane production performance

The impact of NBW addition on methanogenic performance was evaluated in batch anaerobic digestion experiments. As shown in Fig. 2A, NBW significantly enhanced the cumulative methane yield compared to the DW ($P < 0.05$). The cumulative methane production of $O_3$-NBW, Air-NBW, and $O_2$-NBW were 143.0 ± 3.7, 109.0 ± 6.1, and 94.1 ± 3.0 NmL/g-VS, respectively, increased by 87.5%, 43.0%, and 23.4% compared with DW (76.2 ± 3.7 NmL/g-VS). The COD balance analysis also demonstrates that NBs promote methane production (Table S3, Text S1). In the $O_3$-NBW, approximately 28.8% of the total COD was converted to methane, which is higher than DW. The conversion of COD from the solid/liquid to the gas indicates that NBs facilitate the mineralization of organic matter.

The trend of daily methane production was similar for all experimental groups, but the peak methane production appeared earlier and peaked higher in the NBW treatments, while the DW did not show this feature (Fig. 2B). In addition, with the addition of NBW, the peak of daily methane production appeared earlier and the duration of the high methane production state was prolonged, a phenomenon that effectively shortened the stagnation phase at the beginning of the reaction startup (Fig. 2B). Of interest, the $O_3$-NBW showed two distinct peaks in daily methane production, the first peak is presumed to be associated with the rapid degradation and fermentation of readily biodegradable organics, whereas the second peak may be attributed to the subsequent oxidative decomposition of refractory organics induced by $O_3$ NBs to be converted into substances that can be utilized by microorganisms and metabolized for methane production (Lu et al., 2020).

### 3.3. Changes in soluble organic matter and volatile fatty acids

In addition to the methanogenic process, the effect of NBW on the conversion of organic matter in the AD was also reflected in the changes in the content of SCOD and VFA (Fig. 2C-D, Table S4). During days 0–7, SCOD undergoes rapid depletion, indicating that SCOD are rapidly transformed during the early stages of AD. This signifies the presence of active hydrolysis and acidification processes. It is noteworthy that the SCOD removal efficiencies of the reactors in all NBW treatments were higher than DW, with the removal rates of the $O_3$-NBW, Air-NBW, and $O_2$-NBW ranging from 84.6% to 85.8%, which were clearly higher than those of DW (78.7%). Secondly, the addition of NBW promoted the turnover of VFAs. In the early stage of hydrolytic acidification of AD (0–7 d), the accumulation of VFA was higher in the NBW treatments. VFA concentration peaked on the seventh day. Although alkalinity was not directly measured, the pH in the reactor remained between 7.9 and 8.1, indicating no acidification occurred during digestion. After entering the stage of 7–14 d, VFA began to be rapidly consumed, and this process was concomitant with the active period of methanogenesis, and the change was most significant in the $O_3$-NBW. Critically, this rapid turnover process of VFAs is synchronized with methanogenesis, suggesting that NBW drives the methanogenic process by increasing the turnover

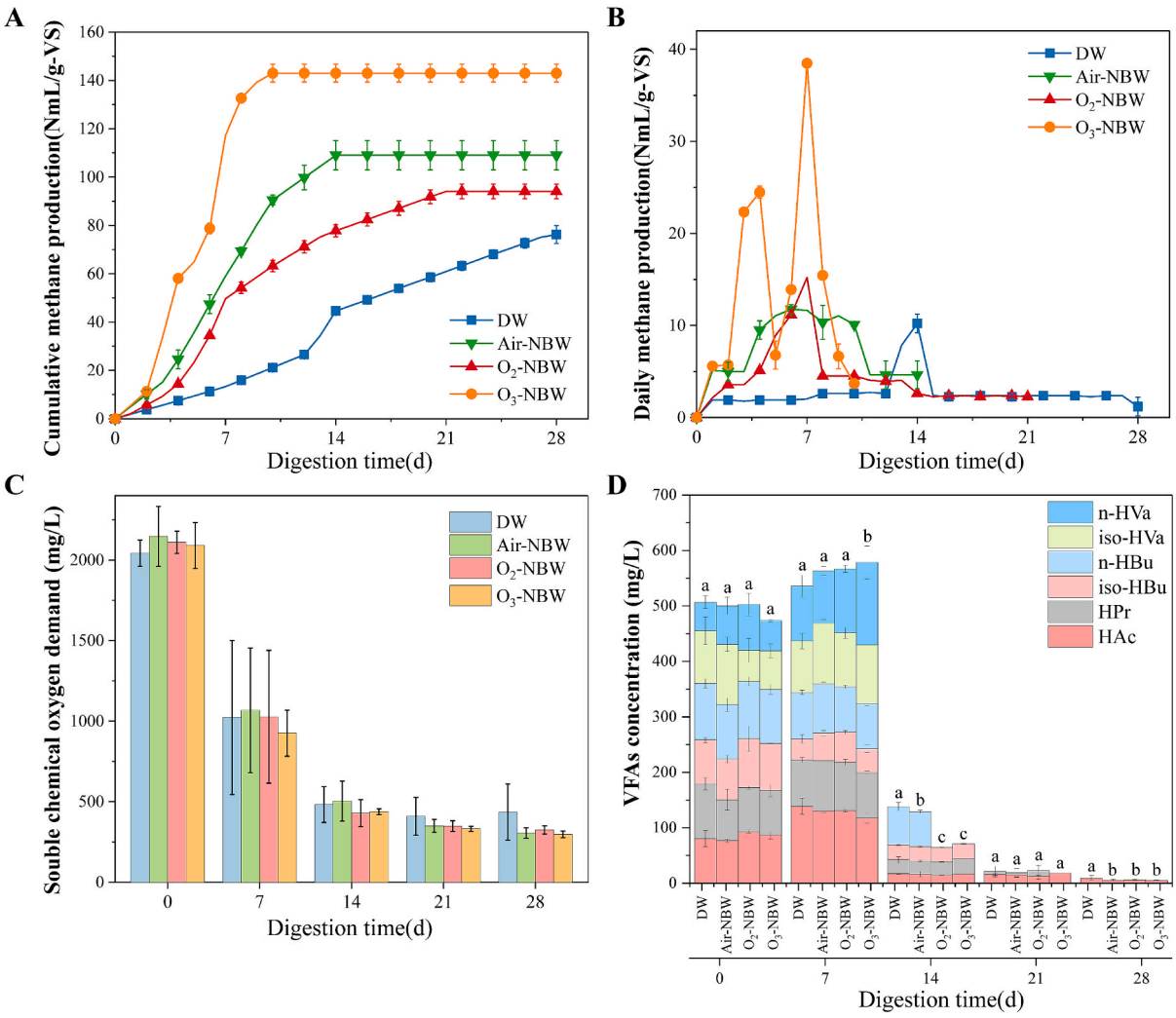



**Fig. 2.** Changes in key performance parameters during anaerobic digestion under different treatments. **A** Cumulative methane production. **B** Daily methane production. **C** Soluble chemical oxygen demand. **D** Composition and concentration of volatile fatty acids (VFAs), including acetate (HAc), propionate (HPr), *n*-butyrate (*n*-HBu), isobutyrate (isoHBu), *n*-valerate (*n*-HVa), and isovalerate (isoHVa). Different lowercase letters in panel D indicate significant differences among treatments (Tukey's HSD, $P < 0.05$). Data represent the mean ± standard deviation (n = 3).

rate of VFAs, enabling more efficient conversion of this key intermediate product to methane (Yang et al., 2020).

In addition, at the late stage of AD (14–28 d), the TAN was higher in NBW treatments than in the control DW (Fig. S1A). The increase in TAN concentration may be associated with the mineralization of nitrogen-containing organic compounds a common process during anaerobic digestion (Yang et al., 2019). It is noteworthy that no ammonia inhibition occurred in all reactors due to low TAN concentrations (less than 1700 mg-N/L), which might be attributed to the low concentration of TS in this study (Fan et al., 2020). Similarly, the pH value in all groups gradually increased after the initial decline due to VFAs converted to methane (Fig. S1B), indicating that the AD process was not inhibited by VFAs accumulation. The concentration of free ammonia nitrogen (FAN) was calculated based on TAN and pH (Text S2), the FAN concentration ranged from 38.3 to 84.8 mg-N/L during digestion (Table S5). On Day 0 of digestion, the highest FAN concentration (84.8 mg-N/L) was observed in the $O_2$-NBW reactor, followed by 79.4 mg-N/L in the $O_3$-NBW reactor on Day 28. Previous studies indicate that FAN concentrations below 150–200 mg-N/L do not usually produce a significant inhibitory effect (Rajagopal et al., 2013; Yenigün and Demirel, 2013). Therefore, in this study, FAN inhibition is unlikely to significantly affect methane production.

### 3.4. Characterization of microbial communities

In order to clarify the correlation between methanogenesis and microbial community structure during AD with the addition of NBW, the microbial community in the system was analyzed by metagenomic sequencing in this study. After quality control, sequence splicing, gene prediction and de-redundancy, a total of 2,862,918 valid genes were acquired. The results of principal component analysis showed that the community distance between NBW treatments and DW was greater on day 28 than on day 14 (Fig. S2B), suggesting that the addition of NBW consistently drove changes in the composition of the microbial community and enriched specific microbial taxa.

At the domain level, the bacteria (94.0–98.4%) and archaea (1.4–5.7%) were the main dominant domains in all groups on day 28 of AD (Fig. S2A), and the proportion of archaea in the NBW treatments were 2.3 (Air-NBW), 2.1 ($O_2$-NBW), and 4.1 ($O_3$-NBW) times that in DW, respectively, showing that methanogens might be enriched. At the phylum level, *Euryarchaeota* is the dominant group, which is consistent with the findings of Fan et al. (2021a). *Euryarchaeota* is recognized as the major methanogenic phylum (Zhang et al., 2020), showed a clear trend of increasing relative abundance in all treatment groups. Whether incubated for 14 d or 28 d, their relative abundance showed a pattern of DW < $O_2$-NBW < Air-NBW < $O_3$-NBW, and this ordering was highly

consistent with the trend of cumulative methane production (Fig. 2A and Fig. 3A). Furthermore, the relative abundance of *Euryarchaeota* in $O_3$-NBW was significantly higher than that in $O_2$-NBW and Air-NBW ($P < 0.05$). This suggests that the addition of $O_3$ NBs may have created a more suitable anaerobic environment, providing stable conditions for the proliferation of methanogenic archaea (Wang et al., 2017). At the same time, $O_3$ NBs may also oxidize complex organic compounds into compounds with lower molecular weights and higher bioavailability, thereby increasing the substrate available to methanogens, which in turn promotes the enrichment of *Euryarchaeota*. At the genus level, the dominant archaeal genus in all samples was *Methanothrix*, with relative abundance ranging from 41.8% to 76.1%, followed by *unclassified_p_Euryarchaeota*, *Methanobacterium* and *Methanosarcina*, with relative abundance ranging from 5.3% to 13.9%, 0.5% to 17.7% and 1.1% to 23.7%, respectively (Fig. 3B). Consistent with findings from previous studies, *Methanobacterium* was also identified as a major genus in the present study (Fan et al., 2024). As a hydrogenotrophic methanogen, *Methanobacterium* primarily utilizes $CO_2$ and $H_2$ as energy sources. Overall, the combined relative abundance of these four dominant methanogenic genera on day 28 increased in the order DW < $O_2$-NBW < Air-NBW < $O_3$-NBW, which was consistent with the observed enhancement in cumulative methane yield.

At the phylum level, the bacterial communities of all treatment groups were dominated by *Proteobacteria* (25.0–57.8%), *Bacteroidetes* (8.8–20.6%), *Chloroflexi* (5.6–31.2%), *Firmicutes* (3.1–14.1%), and *Actinobacteria* (2.2–8.0%) (Fig. 3A). This is consistent with the results of previous studies on microorganisms (Zhou et al., 2024). On day 28, the relative abundances of *Chloroflexi*, *Firmicutes*, *Actinobacteria*, *Planctomycetes*, and *Candidatus_Hydrogenedentes* in all NBW treatments were markedly higher than those observed in DW. Among these taxa, *Chloroflexi*, which are characterized as filamentous multicellular microorganisms, are known to facilitate carbohydrate hydrolysis (X. Wang et al., 2020b). *Firmicutes* can participate in the AD process by secreting extracellular enzymes involved in substrate degradation and VFA production, which in turn provide suitable precursors for the methanogenic process. In addition, *Actinobacteria* and *Candidatus_Hydrogenedentes* have been shown to have the ability to degrade organic matter, including proteins, lipids and polysaccharides (Xiao et al., 2020). Overall, these microorganisms are primarily responsible for driving the hydrolysis and acidification stages of AD. It is worth noting that their abundance was higher in the NBW treatment, suggesting that the introduction of NBs may have created a more favorable microenvironment for the hydrolytic microbial community, thereby accelerating the breakdown of complex organic matter. Furthermore, *Planctomycetes*, which are commonly associated with denitrification processes, showed increased abundance under NBW addition. This enrichment may be related to the elevated availability of nitrogen released from substrate degradation in the presence of NBW, a trend that aligns with the observed variation in TAN concentrations (Fig. S1A) (Liang et al., 2021). Consistent patterns were also observed at the genus level, where NBW addition resulted in pronounced shifts in bacterial community structure during AD (Fig. S3), in agreement with previous studies (Wang et al., 2020a). Compared with the control, *unclassified_d_Bacteroidales*, *unclassified_c_Anaerolineae*, *Lentimicrobium*, and *unclassified_c_Deltaproteobacteria* were notably enriched in the NBW treatments on both day 14 and day 28. *Bacteroidales* is closely related to the process of glucose hydrolysis, and can produce acetic acid and propionic acid through the fermentation of saccharides, and the two are the main metabolic end-products (Qin et al., 2020). *Anaerolineae* is capable of degrading recalcitrant organic compounds (Li et al., 2019). *Lentimicrobium* is a genus of strictly fermentative bacteria capable of converting complex organic matter into VFAs, ethanol and $H_2/CO_2$ (Zhang et al., 2021a).

### 3.5. Metagenomic analysis of methane metabolism

To further elucidate how NBW addition reshapes microbial function to enhance methanogenesis, we analyzed the abundance of genes encoding key enzymes in the three major methanogenic pathways: acetoclastic (M00357), hydrogenotrophic (M00567), and methylotrophic (M00356 and M00563) (Fig. 4, Table S6). The gene abundance of methanogenic pathways was estimated based on KEGG functional modules using TPM normalization. The TPM values of genes associated with each module were summed to represent the relative abundance of the corresponding methanogenic pathway. Overall, the abundance of these methanogenesis-associated genes was significantly higher in NBW treatments than in the DW ($P < 0.001$), indicating a systemic enhancement of the microbial methanogenic potential by NBW (Fig. S4). Notably, $O_3$-NBW exhibited the highest total gene abundance from day

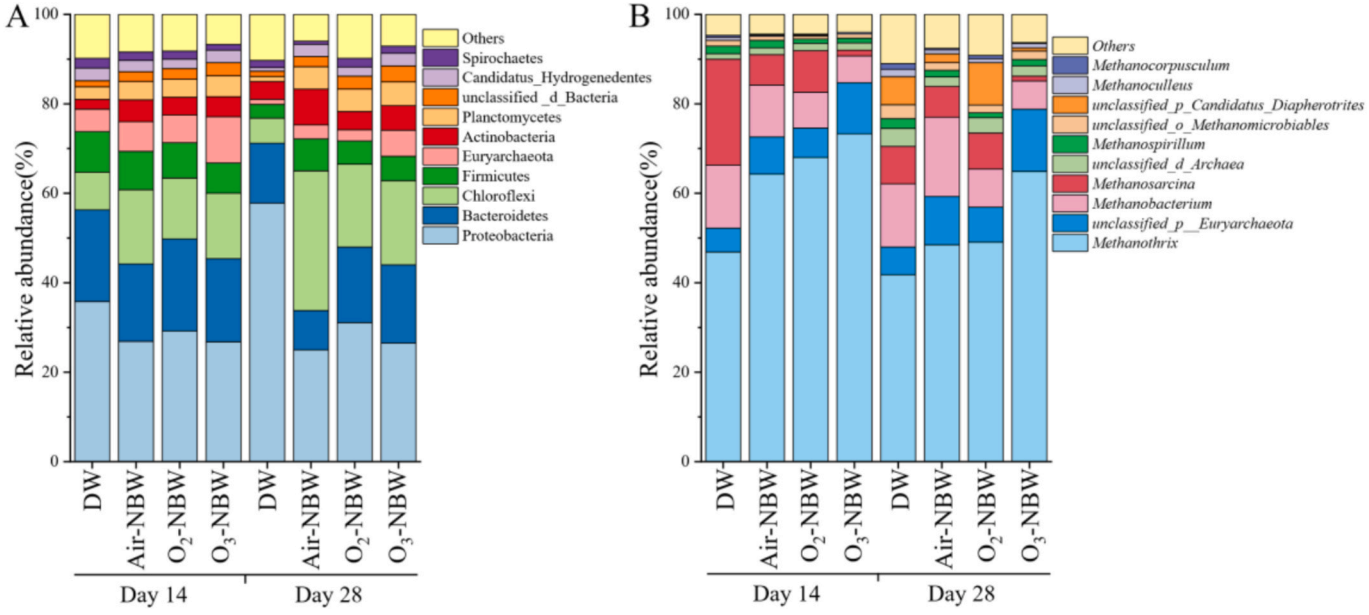



**Fig. 3.** Microbial community composition in different treatment groups during anaerobic digestion. **A** Relative abundance of bacterial communities at the phylum level. **B** Relative abundance of archaeal communities at the genus level. Samples were collected from different treatments at the indicated stages of anaerobic digestion. Taxa with relative abundance < 1% were grouped as "Others".

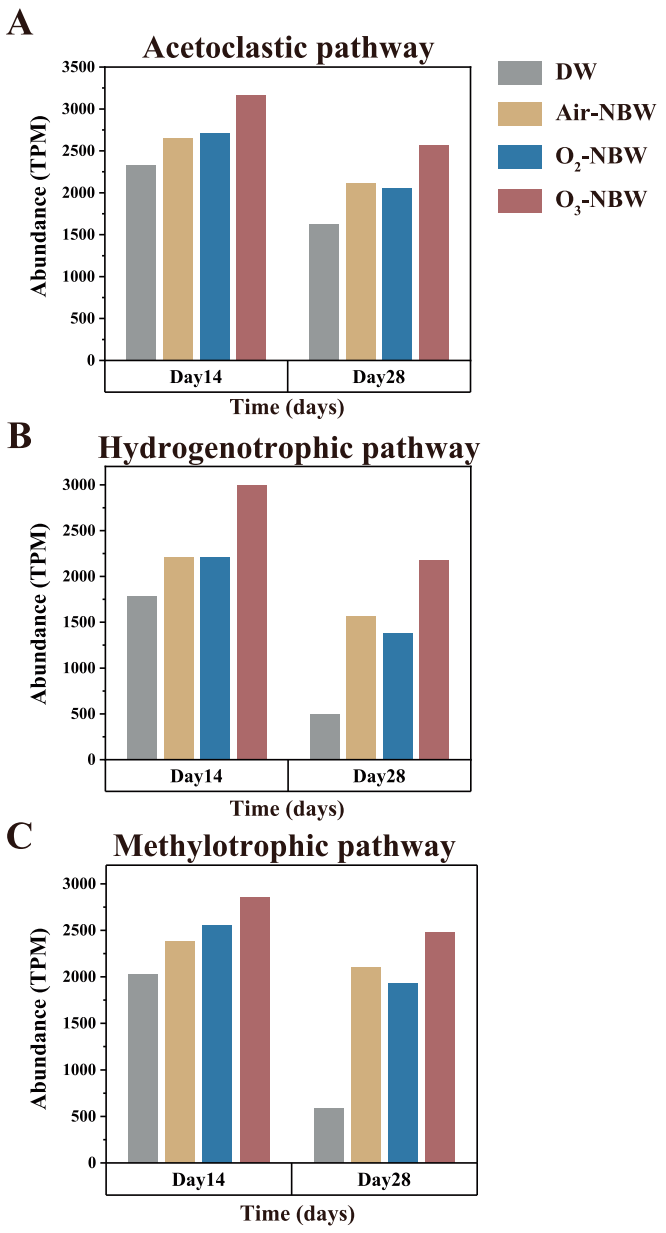



**Fig. 4.** Abundance of genes encoding key enzymes involved in methanogenic pathways in different treatments. **A** Acetoclastic methanogenesis pathway (KEGG module M00357), **B** Hydrogenotrophic methanogenesis pathway (KEGG module M00567), and **C** Methylotrophic methanogenesis pathways (KEGG modules M00356 and M00563). Pathway abundance was estimated by summing TPM-normalized gene abundances for all genes assigned to the corresponding KEGG module. Detailed information on the specific genes included in each module is provided in Table S6.

14 to day 28, a finding that directly correlates with its superior methane production (Fig. 2A). Although a general decrease in gene abundance was observed from day 14 to day 28 across all reactor, the levels in NBW treatments remained consistently higher than in the DW Quantitatively, on day 28, the gene abundances associated with the hydrogenotrophic, methylotrophic, and acetoclastic methanogenic pathways in the NBW treatments were 2.75–4.35-fold, 3.27–4.20-fold, and 1.26–1.59-fold those of the DW control, respectively. Notably, the acetoclastic pathway consistently exhibited the highest total gene abundance in all groups. This is concordant with the microbial community structure, as acetoclastic methanogen (*Methanothrix*) was the dominant archaea in all groups (Fig. 3B) (Ma et al., 2021). These findings collectively indicate that NBW addition did not alter the fundamental metabolic hierarchy but rather systematically enhanced the expression of all major methanogenic pathways within the existing dominant framework.

Further analysis of key enzyme-encoding genes indicated that, within the acetoclastic pathway, NBW treatments exhibited higher abundances of genes encoding *acs* and ACDS compared to the DW. The abundance of key functional genes in the hydrogenotrophic methanogenic pathway was notably higher in NBW treatments than in DW. These genes encode essential enzymes, including formylmethanofuran dehydrogenase subunit A (*fmdA*, EC: 1.2.7.12), formylmethanofuran: tetrahydromethanopterin *N*-formyltransferase (*ftr*, EC: 2.3.1.101), methenyltetrahydromethanopterin cyclohydrolase (*mch*, EC: 3.5.4.27), 5,10-methenyltetrahydromethanopterin hydrogenase (*hmd*, EC: 1.12.98.2), methylenetetrahydromethanopterin dehydrogenase (*mtd*, EC: 1.5.98.1), and 5,10-methylenetetrahydromethanopterin reductase (*mer*, EC: 1.5.98.2). This result was in accordance with the change of hydrogenotrophic methanogens (*Methanoculleus*). Interestingly, in addition to the acetoclastic and hydrogenotrophic pathways, the methylotrophic pathway also played an important role in methane formation in this research. Methylotrophic pathway includes methanol methanogenesis (M00356) and trimethylamine/dimethylamine/methylamine methanogenesis (M00563) (Fig. 5A). With the addition of NBW, the related genes of NBW treatments that encoded coenzyme M methyltransferase (*MtaA* and *MtbA*), 5-hydroxybenzimidazolylcobamide Co-methyltransferase (*MtaB*), and corrinoid protein Co-methyltransferase (*MttB*, *MtbB,* and *MtmB*) were enriched compared with DW on day 28.

In addition, the abundance of functional genes of the generic methanogenic pathway was notably higher in NBW treatments, and the genes coding for key enzymes of methanogenesis all showed an increasing trend in abundance, tetrahydromethanopterin S-methyltransferase subunit A (*mtrA*, EC: 2.1.1.86), methyl-coenzyme M reductase alpha subunit (*mcrA*, EC: 2.8.4.1), heterodisulfide reductase subunit D (*hdrD*, EC: 1.8.98.1), heterodisulfide reductase subunit A (*hdrA2*, EC: 1.8.7.3, 1.8.7.4, 1.8.7.5, and 1.8.7.6).

## 4. Discussion

### *4.1. Enhanced methane production and substrate degradation under nanobubble treatments*

The addition of NBW notably improved the overall efficiency of AD, as evidenced by increased methane production and enhanced organic matter degradation (Fig. 2). Consistent with previous studies reporting that NBs can shorten the start-up phase of AD by accelerating cellulose hydrolysis and promoting the formation of VFAs (Fan et al., 2024; Xu et al., 2025), our results indicate that NBW treatment offers distinct kinetic advantages. Notably, in the $O_3$-NBW, the peak methane yield occurred earliest (Fig. 2B), suggesting that NBs effectively alleviated the initial lag phase in SW digestion, likely by facilitating a faster transition from the hydrolysis stage to the active methane production stage.

The dynamic changes in SCOD and VFAs further reveal this enhanced metabolic coupling. In all NBW treatments, the rapid decline in SCOD during the first week indicates an accelerated rate of complex organic matter degradation (Fig. 2C). Although VFA concentrations rose during the initial days (0–7 days), they subsequently dropped sharply (7–14 days), which coincided with a surge in daily methane production (Fig. 2B, 2D). This synchrony suggests that NBW not only increased the ultimate methane yield but, more importantly, accelerated the turnover of metabolic intermediates. By preventing the excessive accumulation of VFA, NBW maintained a more balanced and resilient anaerobic environment. By the end, VFA levels were similar across all reactors, suggesting that NBs mainly improved conversion rates of intermediates rather than changing the final metabolic endpoints.

A comparative analysis of the treatment of different gases with NBs revealed distinct performance strategies. $O_3$-NBW exhibited two distinct

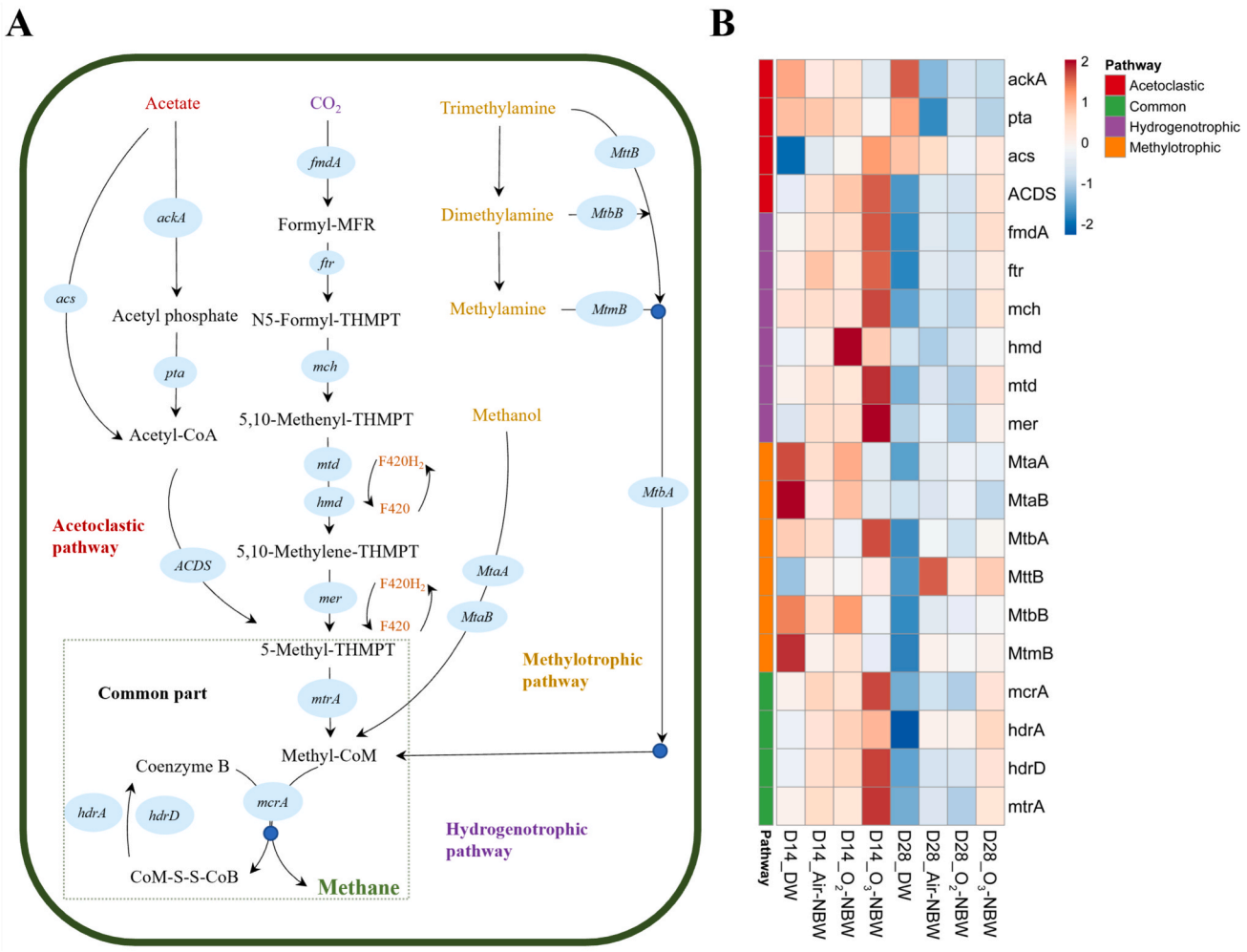



**Fig. 5.** Methane metabolism pathway and abundance of functional genes involved in methanogenesis. **A** Schematic diagram of the three major methanogenic pathways, including acetoclastic, hydrogenotrophic, and methylotrophic methanogenesis, and the key enzymes involved in each pathway. **B** Heatmap for the abundances of functional genes that encoded the key enzymes in different metabolic pathways.

peaks in daily methane production and the highest cumulative yield (Fig. 2A, B). The first peak is presumed to be associated with the rapid degradation and fermentation of readily biodegradable organics, whereas the second peak may be attributed to the subsequent oxidative decomposition of refractory organics induced by $O_3$ NBs to be converted into substances that can be utilized by microorganisms and metabolized for methane production (Lu et al., 2020). It is noteworthy that the final cumulative methane yield of $O_3$-NBW was approximately 1.52 times that of $O_2$-NBW (143.0/94.1). While this ratio coincidentally aligns with the stoichiometric ratio of oxygen atoms in $O_3$ versus $O_2$. The distinct secondary methane peak, unique to the $O_3$-NBW treatment, indicates that the enhanced yield stems from the degradation of a refractory organic fraction that $O_2$-NBW was unable to break down. Therefore, we attribute the higher methane yield to $O_3$'s superior oxidative capacity unlocking an additional carbon source for methanogenesis, not simply a direct stoichiometric conversion. This superior performance highlights that $O_3$ NBs facilitate the start-up phase in the early stages and appear to be more effective than $O_2$ NBs or air NBs in overcoming hydrolysis limitations. Interestingly, $O_2$-NBW produced less methane than Air-NBW (Fig. 2A), which may be because moderate microaeration stimulates facultative microorganisms, but higher dissolved oxygen may shift facultative microbes toward aerobic respiration (Costa et al., 2025), mineralizing organics to $CO_2$ instead of routing carbon through fermentation and methanogenesis (Fu et al., 2023). At the same time, methanogens are oxygen-sensitive, so sustained $O_2$ can directly suppress their activity (Appels et al., 2008).

In summary, these macroscopic performance results confirm that NBW, particularly $O_3$-NBW, act as potent catalysts for AD by accelerating the substrate flux from hydrolysis to methanogenesis. These findings provide a performance-based rationale for further investigation of the effects of chemical pretreatment (Section 4.2) and the changes in microbial communities driven by these unique NBW treatments (Section 4.3).

### 4.2. *Mechanism of enhanced hydrolysis and acidification by nanobubble water*

The significant increase in methane yield and COD removal rate indicates that NBs promote the AD process, particularly by enhancing hydrolysis and acidification, which are typically considered rate-limiting steps. Under various treatment conditions, $O_3$-NBW demonstrated a more pronounced promotional effect. This is likely due primarily to the strong oxidizing capacity of ozone and the reactive oxygen species potentially generated during the collapse of NBs. The high redox potential of ozone, along with the hydroxyl radicals (·OH) released during NB collapse (Varoutoglou et al., 2025), aids in breaking down refractory chemical bonds in complex organic compounds that remain stable under conventional anaerobic conditions or when exposed to oxygen (Koundle et al., 2025). This in-situ chemical pretreatment effectively converts non-degradable components into easily oxidizable

substrates, enhancing substrate bioavailability, which also explains the unique secondary methane peak observed in the $O_3$-NBW reactor.

In addition to chemical oxidation, the physical properties of NBs may also contribute to the sustained transformation of the substrate. Previous studies have shown that NBs can remain stable in water for extended periods under certain conditions (Ji et al., 2020; Liu et al., 2016), although the mechanisms underlying this stability are still under investigation. It is generally believed that the stability of NBs is closely related to their surface charge, interfacial adsorption layers, and the composition of ions and organic matter in the solution. Therefore, in complex aqueous systems such as wastewater, the stability of NBs is typically influenced by ionic strength, and dissolved organic matter. Although we attempted to monitor NB characteristics during AD, the high turbidity and complex particulate/biological background of the reactor matrix caused severe analytical interference, preventing reliable real-time determination of NB size and concentration. In this study, methane production and substrate degradation showed an upward trend (as shown in Section 4.1), while the microbial community structure also exhibited certain functional adaptations (as described in Section 4.3). These phenomena suggest, to some extent, that the introduction of NBs produces sustained functional effects in the AD system. Their nanoscale size results in extremely low buoyancy, significantly prolonging their residence time within the reactor (Favvas et al., 2021). Additionally, the typically negative charge on the surface of NBs (high zeta potential) effectively inhibits coalescence and premature dissolution through electrostatic repulsion (Wang and Lu, 2025). These physicochemical characteristics may contribute to the persistence of NBs after introduction into complex systems However, the in situ direct monitoring of NB size and concentration in AD under conditions of high suspended solids concentration remains a significant technical challenge in this field (Chuenchart et al., 2021; Fan et al., 2021b; Jia et al., 2023; Wang et al., 2023). Therefore, the size evolution and concentration changes of NBs throughout the entire AD process, as well as their relationship with enhanced treatment effects, require further investigation.

Unlike macro-bubbles, NBs possess an extremely high specific surface area and mass transfer efficiency, which significantly increases the frequency of contact between enzymes and substrates (Wang et al., 2026). $O_3$ NBs also tend to carry a strongly negative zeta potential, so electrostatic repulsion limits coalescence and helps them remain stable (Singh et al., 2024). According to the literature, thanks to this stability, $O_3$ NBs in such systems may have contributed to maintain dissolved oxygen at a relatively low level of approximately 0.2–1.0 mg/L, thereby creating a sustained micro-oxidative environment (Tsai et al., 2025). This environment provides optimal conditions for facultative hydrolytic bacteria, allowing them to function normally over time without adversely affecting strict methanogens.

More fundamentally, the high interfacial free energy of the negatively charged surfaces of NBs reduces the surface tension of the wastewater matrix. This physical modification alters the surface properties of particulate organic matter, making it more susceptible to microbial attachment (Yasui et al., 2019). By promoting bacterial attachment to solid surfaces, NBW facilitates the formation of robust biofilms and enhances the spatial proximity between hydrolytic enzymes and complex substrates. These combined physical and chemical mechanisms collectively accelerate the carbon transfer from particulate organic matter to volatile fatty acids, thereby supporting a significant improvement in overall digestion performance.

### *4.3. Impact of nanobubble water on microbial community and methanogenic pathways*

In addition to improved substrate conversion, NBW treatment was accompanied by changes in microbial community structure and methanogenesis-related functional genes. These changes showed good correlation with increased methane production, suggesting that NBW treatment may have further influenced the methane production process within the system. However, it should be noted that these biological responses more likely reflect adaptive adjustments resulting from improved substrate availability and changes in the reaction microenvironment.

In this study, we found that the abundance of certain genes associated with electron transfer-related structures (such as *PilA* and *OmcS*) decreased in the NBW treatments (Fig. S5), while methane production increased simultaneously. This phenomenon suggests that NBW treatment may have altered the electron transport strategy within the system and reduced the genetic investment in certain energy-intensive biological conductive structures. We speculate that the interfacial electrical properties and surface characteristics of NBs may have influenced the spatial contact or electron transfer resistance among symbiotic microorganisms to some extent, thereby reducing the microbial community's investment in structures such as conductive pili and redirecting more metabolic resources toward the methanogenesis process. This phenomenon is consistent with observations in systems where inorganic conductive materials were added (Worajo et al., 2025).

Changes in the microbial community provide indirect support for the above hypothesis. The significant enrichment of *Methanothrix* during NBW treatment suggests that a microbial community structure more conducive to efficient methane production may have formed (Fig. 3B). Because *Methanothrix* is a typical methanogenic archaeon, it is able to receive electrons directly through DIET and reduce $CO_2$ to methane (Zhao et al., 2020). Liu et al. similarly found that NBs promote the enrichment of *Syntrophomonas*, a bacterium which, like *Methanothrix*, is a typical DIET-functioning microorganism (Liu et al., 2024). These community changes suggest that NBW treatment is associated with community restructuring related to electron transfer.

NBW treatment was also accompanied by an enhancement of multiple methane-producing pathways, shifting from the predominant acetoclastic methanogenic pathway to a synergistic, multi-pathway metabolic network. Although acetoclastic methanogenic pathway remains the primary pathway for methane production, the gene abundance of key enzymes in the hydrogenotrophic and methylotrophic pathways increased significantly following NBW treatment. This shift in metabolic pattern may be related to enhanced substrate availability and improved mass transfer efficiency.

Particularly under $O_3$-NBW, stronger oxidative effects may promote the conversion of some recalcitrant organic compounds into small-molecule organic acids, fermentable sugars, and other intermediate metabolites, thereby providing a richer substrate source for various methanogenic pathways (Auma Omondi and Aluda Kegode, 2023; Fan et al., 2024; Patel et al., 2021) At the same time, NBs may further enhance the synergy among multiple methane-producing pathways by improving gas–liquid mass transfer and the local hydrogen transfer environment (Chuenchart et al., 2021). This metabolic flexibility is reflected to some extent in the microbial community structure. The abundance of the *Methanosarcina* serves as an example; its ability to utilize acetate, hydrogen/carbon dioxide, and methyl compounds demonstrates the community's adaptability to multi-pathway strategies.

It is worth noting that, although these results support an association between NBW treatment and the restructuring of the methanogenic network, this study cannot yet strictly distinguish whether this biological response stems from the direct effects of NBs on microorganisms or from indirect effects resulting from the physicochemical transformation of substrates. Combined with the results in Section 4.2, the current evidence more directly supports the hypothesis that $O_3$-NBW first improves substrate availability by enhancing the transformation of complex organic matter, while changes in microbial communities and functional genes are more likely to be subsequent responses to these substrate and environmental changes. Therefore, whether NBs directly stimulate microbial metabolism still requires further verification through more discriminatory experimental designs, such as establishing gas-specific NBW treatment systems containing only wastewater or only inoculated sludge, respectively, to distinguish between substrate effects

and direct microbial responses.

Overall, the microbial community restructuring and enhanced methanogenesis observed in this study suggest that NBs not only influence substrate transformation but also alter methane metabolism in AD systems. This effect stems from the coupling of physicochemical interactions with subsequent microbial responses, rather than a single, direct microbial stimulation mechanism.

### 4.4. Practical Implications and Future Perspectives

The findings of this study provide key insights into the application of NB technology in the sustainable management of SW, demonstrating the potential of NBs to transform the conventional AD process into a highly efficient resource recovery process. Integrating $O_3$-NBW into AD systems offers significant economic and environmental benefits. By promoting efficient methanogenesis through enhanced hydrolysis, $O_3$-NBW nearly doubles the energy recovered per unit of reactor volume. This improvement directly enhances the project's economic viability by increasing biogas revenue and potentially reducing net treatment costs through reduced sludge treatment and shorter retention times. These findings are highly aligned with global strategic goals for resource recovery and carbon neutrality in the agricultural sector.

Beyond organics, swine wastewater often carries residual antibiotics and disinfectants that suppress AD microbiology. $O_3$ NBs may also reduce this inhibition by oxidizing such biotoxic compounds in situ, improving effluent safety, easing the load on downstream processes, and lowering ecological risks after discharge. Although $O_3$ NBs technology shows good potential for application, its practical engineering application still faces many challenges. Future research should focus on advancing the scaled-up development of NBs generation systems and exploring their long-term operational stability under different climatic conditions and feedstock components. In addition, in-depth study of the synergistic effect of NBs and other process enhancements, including inoculation of functional microbial for biological enhancement, combined with solar thermal technology, can open up a new direction for the research and development of intelligent, near-zero emission sewage treatment system for livestock and poultry breeding industry.

## 5. Conclusion

This study shows that adding nanobubble water to anaerobic digestion of swine wastewater markedly increases methane recovery, with $O_3$-NBW giving the greatest improvement (87.5%). The process shifts from purely biological conversion to a synergistic mechanism in which nanobubbles enhance hydrolysis and change the electron transfer strategy. This is supported by the enrichment of *Methanothrix* and the downregulation of the energy-demanding *pilA* gene. Because the experiment lasted 28 days, longer runs (60–90 days) are required to evaluate microbial community stability and confirm whether $O_3$-NBW advantages persist.

From an engineering viewpoint, $O_3$-NBW can provide in situ pretreatment while stimulating metabolic activity, potentially shortening hydraulic retention time, reducing reactor footprint, and improving on-farm energy self-sufficiency via higher biogas yields. However, scale-up faces challenges. $O_3$ generation is energy-intensive and needs specialized equipment, raising capital and operating costs. Nanobubble production via high-pressure dissolution requires robust pumps and accurate pressure control, adding further energy demand. In addition, while ozone can break down complex organics, excessive dosing may create oxidative stress and inhibit strict anaerobic methanogens during long-term continuous operation.

Future studies should conduct techno-economic analysis and life-cycle assessment to determine whether extra methane production offsets the energy required for ozone and nanobubble generation. Work is also needed to test electron transfer stability in continuous reactors and explore synergy between nanobubbles and conductive carbon materials (e.g., biochar), thereby further improving efficiency and stability of NB-AD systems.

## CRediT authorship contribution statement

**Xiaoxuan Liu:** Writing – original draft. **Huacai Wang:** Writing – original draft. **Sining Zhou:** Methodology. **Yawen Xie:** Methodology. **Jinglin Wang:** Data curation. **Xu Wang:** Data curation. **Shengjun Xu:** Writing – review & editing. **Lei Wang:** Methodology. **Cancan Jiang:** Writing – review & editing, Supervision. **Xuliang Zhuang:** Writing – review & editing, Funding acquisition.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.biortech.2026.134692.

## Data availability

Data will be made available on request.

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